Siren: Tracks & Tech

Siren turns your car into an ambulance: speak your symptoms, go to the ER that has the right doctors and beds right now, and the ER is ready before you arrive. Ambulance crews and families are connected on the same live system.

Tracks we're competing in

Track	Type	Our angle in one line
SpacetimeDB	Sponsor track	SpacetimeDB is Siren's entire live backend: the shared state that phones, the nurse board, the ambulance tablet and our server all coordinate through.
Cartesia	Sponsor track	Voice is the product: patients and paramedics talk instead of typing, and Siren talks back.
Anthropic (Claude)	Sponsor track	Claude does the clinical thinking and runs a safe, well-limited agent, and we measured it against real nurses.
Microsoft: Built for the Real World	Track	Built around a real problem (a teammate's 40-minute unwatched chest-pain wait) and real ER workflows.
Meta: Bringing People Closer Together with AI	Track	Siren connects people at the moment they feel most alone.

Why we fit each track

SpacetimeDB · Sponsor track

SpacetimeDB is Siren's entire live backend: the shared state that phones, the nurse board, the ambulance tablet and our server all coordinate through.

- 13 tables and ~25 reducers with guards (server-only, staff-only, phone ownership).
- **Private tables + per-user views** (my_patients, my_vitals, my_alerts, my_offers, my_agent_actions): a phone or ambulance tablet sees only its own patient; staff see everything.
- Safety rules enforced **in reducers** (risk only escalates; no redirecting critical or ambulance patients), so no client can bypass them.
- Live everywhere: board, phone, family page and crew tablet update instantly via the React SDK (useTable, useReducer); the Python server writes through the HTTP API.
- Schema migrations with defaults; hosted on Maincloud (database frontdoor-w3zl2).

Cartesia · Sponsor track

Voice is the product: patients and paramedics talk instead of typing, and Siren talks back.

• **Ink** (ink-whisper) speech-to-text for check-ins, waiting-room answers and paramedic reports, in English, Spanish, Chinese, Vietnamese and Tagalog.
• **Sonic** (sonic-3.6) speaks replies and check-in questions in the patient's language, and reads hospital replies aloud to the ambulance crew ("Bed held for your patient") so their hands stay free.
• Automatic voice check-ins every 2 minutes; speech-to-text glitches on noise are filtered so they never page a nurse; two unclear answers in a row still alert one.
• Accessibility: people who don't speak English, or can't type while scared or driving, can still check in.

Anthropic (Claude) · Sponsor track

Claude does the clinical thinking and runs a safe, well-limited agent, and we measured it against real nurses.

• **Triage** (claude-opus-5-5, structured outputs): ESI urgency, red flags, department needed, what the ER should prepare; extracts vitals the paramedic says out loud.
• **Measured:** on real ED triage records (MIMIC-IV-ED), the ESI-algorithm prompt caught **96% of the nurses' high-risk patients** (48 of 50) on a held-out test, up from 84%.
• **Charge-Nurse Assistant** (tool use, validated in code): pages departments, holds beds, posts checklists by itself; assigning beds or diverting patients needs a nurse's Approve; it has no tool to lower urgency or discharge anyone.
• Safeguards: deterministic red-flag rules can only raise urgency; rule-based fallback if the API fails; never tells a patient a diagnosis.

Microsoft: Built for the Real World · Track

Built around a real problem (a teammate's 40-minute unwatched chest-pain wait) and real ER workflows.

• **Users:** the charge nurse who owns the board, walk-in patients, ambulance crews, and families.
• **Measurable:** alert → "I'm on it" logged with nurse and time; beds-short warning 30 minutes ahead; department gaps visible before arrival.
• **Responsible:** private patient data, humans approve every bed and diversion decision, EMTALA respected (we never redirect anyone who has arrived), family link shows location and status only.
• **Path to production:** hospital SSO instead of a PIN; bed and on-call data from hospital systems (HL7/FHIR); ambulance vitals from cardiac-monitor feeds.

Meta: Bringing People Closer Together with AI · Track

Siren connects people at the moment they feel most alone.

• **Patient ↔ nurse before arrival:** the ER knows who's coming and what they need.
• **Paramedic ↔ hospital:** the crew hears the hospital's reply while staying with the patient.
• **Across languages:** patients speak five languages and are answered in their own; nurses read English.
• **Patient ↔ family:** a live link shows where the patient is and when they've reached a bed.

Tech stack

Built with: spacetimedb · cartesia · claude · anthropic · python · fastapi · react · typescript · vite · leaflet · openstreetmap · osrm · apify · docker · render · mimic-iv-ed

Layer	Technology	What we use it for
Real-time database	**SpacetimeDB** (Maincloud), TypeScript module	All live state: patients, vitals, alerts, beds, departments, nearby ERs, agent actions; per-user views for privacy
AI	**Claude** claude-opus-5-5 (Anthropic Python SDK)	Triage, department routing, prep items, vitals extraction, waiting-room follow-ups, Charge-Nurse Assistant agent
Voice	**Cartesia** Ink (ink-whisper) + Sonic (sonic-3.6)	Speech-to-text and spoken replies in 5 languages
Server	**Python**, **FastAPI**, Uvicorn, httpx, Pydantic	API, ER matching, vitals simulator, bed forecast, agent loop
Web app	**React 19**, **TypeScript**, **Vite**	Patient phone page, nurse board, ambulance tablet, family page
Maps & routing	**Leaflet** + OpenStreetMap tiles, **OSRM**	Live map of incoming patients; real drive times and ETAs
Location	Browser Geolocation API	Finds the patient or ambulance; no third-party location service
Deployment	Docker, **Render**	One container serves the website and the API
Data collection	**Apify**	Scraped ER entrance locations (Google Maps)

Data sources: what's real, what's simulated

Data	Source	Real or simulated
ER locations	Apify Google Maps scrape	Real
Yearly ER visits per hospital	California **HCAI** (2025)	Real
Licensed bed types (ICU, cardiac, burn, pediatric)	California **CDPH**	Real
Hourly arrival pattern, vital-sign trajectories, triage test set	**MIMIC-IV-ED** Demo (de-identified real ER stays)	Real, replayed
Beds free right now, doctors on call, equipment status, waits	Generated by our server	Simulated (only inside hospital systems; would come from HL7/FHIR)